

Research Assignment 1: Foundation of Data Analytics and Data Science.

SECTION A: DATA FUNDAMENTALS.

1. What is a Database

A database is a system used to store and collect data. The data that is stored can be in the form of numbers, images as well as videos.

Databases are essential for the following reasons:

- (1) Efficient Scaling - It is the ability to manage a lot of data which can be scaled into millions, billions and even more. In order to be able to store these huge amounts of data one needs databases to manage it. Example can be social media platforms.
- (2) Data Integrity - It refers to the build in rules that is built in these databases to be able to maintain data consistency. Example of data integrity is to make sure that a Product table have a Product ID.
- (3) Data Security - It refers to the ability of database to be able to

support the privacy and compliance
that of data. Example access control which
is often associated with
allow different users that have different
level of access. ie a supervisor vs Controller.

- 4) Data Analytics. It is a software
used to analyse data in order to
to identify trends and patterns in
order which will make it possible to
make predictions. Example determine
which product generate the most sales.

Two Real Examples where database are
used:

- (1) Social media Platforms.
- (2) Stock Exchange

(2)(a) WHAT IS A Data Warehouse

— It is the ability to bring
all the source data from various
sources which are stored in these
data warehouses which are used to
optimize these data for query and
analysing.

(b) How does it differ from a regular
database?

A data warehouse is used to store
a lot of historical data and information
for analytical purpose.

(c) Why would a Company use a data warehouse?

The reason why a Company use a data warehouse is to being able to optimized budgeting and speed, better quality of data, as well as for security purposes.

(3)
(4) What is a Data Lake? Explain the Concept and describes how it differs from a data warehouse?

A data lake is centralized which allows it to store large amounts of data in its original form. These data can be structured, semi-structured to even unstructured data. Data warehouse is where the data is clean and prepared in order to do analysis as well as for business intelligence. As oppose to data lakes which store large amount of raw data at a low cost.

A business would prefer data lakes over data warehouses because data lakes offer lower and flexible storage cost and are faster.

4) (a) WHAT IS A DATA MART?

A data mart is a subsection of a data warehouse that focus on a specific department or subject area. It can assist with cost reduction, improve the efficiency of the team and help with better decisions that are made in a business.

(b) How DOES IT RELATE TO A DATA WAREHOUSE and who typically use it?

(1) Scope : A data warehouse has more a broader scope as it cover the entire enterprise as while a data mart are only responsible for a specific department or subject.

(2) Relationship : A data mart is a subsection of the broader data warehouse

WHO use it.

- Sales, marketing department use them to access specialized data
- Managers also use them to monitor KPI of a specific department.

5) Explain the difference between structured, semi-structured and unstructured data. and give an example of each.

(1) Structured data: It is data that is well organized. An example the financial transaction at the bank, customer database and Point of sales.

(2) Semi-structured data: It is data that sits in between structured and unstructured data. It is used in

industries that handle large amounts of data. Example social media feeds

(3) Unstructured data: It does not operate according to a specific data model which makes it difficult to be organized. Example video/audio analysis

(b)(i) WHAT IS A DATA MODELING.

Data modeling refers to the process to create a structured representation of data. The goal of data modeling is to simplify complex data so one can easily understand and analyse it.

(b) WHY IS IT IMPORTANT WHEN DESIGNING A SYSTEM THAT WORKS WITH DATA. ?

- It ensure that the system is efficient, ~~easy~~ Scalable and secure.
- It also ensure data integrity and improve performance

(6) Describe two types of data models?

(1) Relational Data model.

- It is where data is grouped in tables which are then organised in rows and columns.

(2) Network data models

- It organises data in a more flexible series of relationships. It allows for more flexibility and complexity in the relationship between data points.

(7) Data mining is use of statistical analysis to solve and uncover patterns in large datasets. Example of data mining fraud detection in the banking industry, as well as marketing campaigns.

(8) A data base management system is an interface between users, applications and data which allow for retrieving and management of structured and unstructured data.

(i) MySQL: It is very reliable and easy to use.

(ii) Oracle Database: It is used for large-scale applications which require high performance.

- (3) Microsoft SQL Server: It is used for data integration and analysing of data.
- (4) PostgreSQL: It is known for its compliance standards and performance.
- (9) It is a process where raw data is collected which is then cleaned, converted and then loaded in a data warehouse for analysis.

- (1) Extract: Allow to extract raw data from multiple sources.
- (2) Transform: It is the process to clean and filter data to ensure consistency.
- (3) Load: Import high-quality data into data warehouses for analysis.

Importance

- Improve decision making
- Efficient operations
- Compliance and security

- (10) (1) Parquet: It is ideal for large datasets due to superior compression and fast read times.
- (2) JSON: It is used in API, document database and semi-structured data.
- (3) CSV: Ideal for small data exports.
- (4) Avro: It is ideal for high speed writing.

Xm: Use an older web services,
configuration and system integration.

Returne dist.

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SECTION B : Artificial Intelligence and Machine Learning.

- (11) Artificial intelligence is technology that enables computers and machines to simulate human learning, comprehension, problem solving and decision making.

NARROW AI - It is designed to perform a specific task or a limited range of tasks.

- It is typically built to excel in one domain and can perform superhuman performance in that area but lack general understanding or awareness.

Example Facial Recognition

General AI : It is designed to generate new content.

- It works by learning from vast datasets to understand patterns and structures which allows them to produce new outcomes.

Example ChatGPT

- (12) It is a subsection of artificial intelligence focused on algorithms that can learn the patterns of training data from which enables them to make accurate inferences about new data.

- (1) Supervised - Learning from labelled data
 - (2) Unsupervised - Discovering patterns in unlabelled data
 - (3) Reinforcement - Learning through interactions with an environment
- (13) Deep learning powers most state of the art artificial intelligence today from computers, generative AI to self driving cars and robots.

It relates to artificial intelligence and machine learning because it is a specialized subset of machine learning which is a subset of artificial intelligence.

Neural networks are computing systems inspired by the human brain which was designed to recognize patterns and solve complex problems by stimulating interconnected nodes. They operate by passing data through layers - input, output and hidden - adjusting internal weights through training to map inputs to accurate outputs, frequently used in image recognition and AI.

- 14) It ~~was~~ uses deep learning - specifically artificial neural networks to enable computers to understand, interpret and generate human language often using transformer language models like GPT and BERT.
- Example: Virtual Assistant and Chatbots

15) They are a category of deep learning models trained on immense amounts of data, making them capable of understanding and generating natural language and other type of content — to perform a wide range of tasks.

GPT-4 (OpenAI) The models that run the ChatGPT service.

Gemini (Google). This is a family of natively multimodal models, including Pro and Flash made by Google Deepmind.

16) Generative AI is a type of artificial intelligence that creates new content, including text, images, codes and videos by learning patterns from massive datasets.

It works on high-level by using neural networks to analyze vast datasets, learning the underlying patterns, structures, and relationships within that data.

17) ~~It~~ Training dataset is dataset used to teach ~~artificial intelligence~~ machine learning models to recognize patterns, make predictions or generate content.

If training data is biased or incorrect is a major risk in machine learning.

leading to models that learn, amplify and perpetuate those flaws.

18) It is field of artificial intelligence that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs, allowing them to take action or make recommendations.

- (1) Autonomous Vehicles
- (2) Medical Image Diagnosis

19) API is a set of rules that allows different software applications to communicate and exchange data with each other.

It is important in modern software and data systems because of its

- (1) Efficiency and Cost saving.
- (2) Scalability
- (3) Improves security

20) API key is a unique alphanumeric string used to identify and authenticate an application.

Its purpose is to identify and authenticate software making a request to the API.

- It can be updated by the following:
- (1) Implement Restriction
 - (2) Rotate Key Regular
 - (3) Enable Secret Scanning

When it is exposed it can lead to:

- (1) Financial loss
- (2) Service Disruption
- (3) As well as reputational Damage.

21) Rest API is the application programming interface that conforms to the design principles of the representational state transfer architectural style, a style used to connect distributed hypermedia systems.

- (1) GET: Used to retrieve data from the server
- (2) POST: Used to create new resource
- (3) PUT: Used to replace update or replace an existing resource.
- (4) Delete: Used to remove the resource from the server.

22)

22) JavaScript Object Notation. is an open data interchange format that is both human and machine-readable.

```

{
  "name": "John Doe",
  "age": 30,
  "isStudent": false,
  "address": {
    "city": "New York",
    "zipcode": 10001
  },
  "hobbies": ["reading", "coding", "hiking"],
  "referredBy": null
}

```

23) A prompt engineer is a specialist who designs, refine and optimize text prompts to guide generative AI models toward producing accurate, relevant and high-quality output.

Poor Prompt

- Write a social media post about our new coffee shop.

Improve Prompt

- Act as a professional copy writer. Write an Instagram caption announcing the grand opening of a eco-friendly coffee shop.

24) The Map function is a fundamental tool used across programming, mathematics and spreadsheets to transform a collection of items by applying the same rule or functions to each one.

Example

Python: `map(str.upper, ["apple", "banana"])`

Javascript: `()`

25) Cloud Computing is the on-demand delivery of IT resources - such as servers, storage, databases and software over the Internet with pay-as-you-go pricing.

(1) IaaS: (Infrastructure as a Service)
: Provider fundamental computing resources - virtual servers, storage and networking. over the internet.

Example: Amazon Web Services

(Platform as a Service)

(2) PaaS: Provides a cloud-based environment for developers to build, test, and deploy applications without managing the underlying infrastructure.

Example: Google App Engine

(3) SaaS (Software as a Service): Provides fully functional software applications over the internet usually via subscription over the internet.

Example: Microsoft 365

26) A Version Control system is software that tracks and manages changes to files and code over time, creating a history of modifications.

Git is a free open-source distributed version control system (VCS) that tracks every change made to a project file over time.

Data analyst use Git to manage the complexity of modern, code-heavy analytical projects

27) Jupyter Notebook is an open-source browser-based application used to create and share documents containing live code, visualization and narrative text.

It is used by data scientists by allowing users to run code in small chunks, document steps and immediately visualize results.

It is commonly used with Python programming language

28) ~~Python is a programming language used in web applications. data science~~

Python is widely used in web applications as well as software development.

(1) Numpy: It provides support for large, multi-dimensional arrays and matrices along with a collection of high-level, mathematical functions to operate on these structures.

(2) Pandas: Allows developers to handle tabular data with ease

(3) ~~Mat~~ Matplotlib: It is used to create a wide variety of static, animated, and interactive plots.

29) Data privacy is the right of individuals to control how their personal data such as identifiers, behaviour or sensitive information is collected, used, shared and stored by organisations.

Data privacy is important because it protects personal and financial information from unauthorized access, misuse, and theft directly reducing the risk of identity theft and fraud.

The General data Protection Regulation is a comprehensive European Union law, effective May 25, 2018 that strengthens individuals' rights over personal data and mandates strict privacy standards for handling, storing, and transferring information.

Countries that apply General data Protection Regulation is Ireland, Liechtenstein and Norway.

30) Data ethics is a branch of ethics that evaluates the moral implications of data collection, processing, sharing and usage particularly regarding AI and algorithms.

(1) Algorithmic Hiring Bias (Amazon Recruitment Tool)

— Amazon developed an AI-driven tool to scan resumes and rank job applicants.

The model was trained on resumes submitted to the company over a 10 year period, a time during which the tech industry was predominantly male.

Problem: Because the historical data reflected societal biases, the AI learned that male candidates were preferred. It actively downgraded applicants containing words like women and penalized graduates from women colleges. This creates a discriminatory feedback loop that systematically excludes qualified individuals based on gender rather than merit.

(2) Large - Scale Facial Recognition Scraping.

(Clearview AI)

→ Clearview AI Scraped billions of photos from social media profiles without the knowledge or consent of the users. This database was then sold to law enforcement agencies to identify individuals biometrically.

Problem: This ~~is~~ violates the fundamental right to privacy by repurposing personal photos taken from social interaction into a surveillance tool. Furthermore because facial recognition technology often has lower accuracy rates for people of color, this database risk facilitating wrongful arrest and invasive unchecked surveillance of marginalized communities.

(3) Exploitation of Personal Data for Political Manipulation. (Cambridge Analytica)

A third-party application on Facebook "improperly harvested" data from over 87 million users without their explicit consent. This data combined with psychological profiling was used to create targeted AI-driven advertising designed to influence vote behaviour in political campaigns.

Problem: This constitutes a massive breach of user trust; consent and data privacy. It demonstrates the dangerous misuse of data for social manipulation, undermining democratic processes by targeting users with fabricated or emotionally manipulative content without their knowledge.

(31) Data governance is a structured framework of policies, processes, role and technology used to manage an organization's data; ensuring it is secure, high-quality and usable throughout its lifecycle.

- (1) Policies and Standards: Rules defining how data is collected, stored and used.
- (2) Roles and Responsibilities: Stewardship such as appointing data stewards to enforce rules.
- (3) Security and Privacy: Protecting sensitive data and ensuring regulatory compliance.

4) Data Quality: Ensuring data is accurate, consistent and reliable.

(32) Bias in AI also known as algorithmic or machine learning bias, occurs when an AI system produces systematically skewed, unfair or inaccurate results often reflecting human prejudices, societal inequalities or flawed training data.

Example:

(1) Racial Bias in Healthcare Algorithms.

A widespread algorithm used in US hospitals to predict which patients need additional medical care was found to be biased against Black patients. The model used healthcare cost history as a proxy for medical need, but because of systemic inequalities, less money was historically spent on Black patients, leading the AI to conclude they were healthier than they actually were.

Consequence: Black patients with severe illnesses were denied access to necessary additional care compared to white patients with the same condition.

(2) Facial Recognition Accuracy Disparities. Studies by MIT Media Lab demonstrated that commercial facial recognition systems had drastically higher error rates for darker-skinned

featureless faces

Consequence: Increased false ~~front~~ positives in surveillance, leading to wrongful arrest of innocent people, particularly people of color.

33) DATA

Analyst (Interpreter)

- Focus on analysing historical data to identify trends and inform business decisions

Data Scientist (Modeler)

- Uses statistical analysis and machine learning to forecast future actions.

Data Engineer

(Builder)

- Focus on data infrastructure, pipelines and warehousing

AI Engineer (Architect)

- Builds, deploys and manages intelligent systems that act autonomously.

(34) Business intelligence is a technology driven process that analyzes historical and current data to help organizations make informed, strategic decisions.

BI Tools:

- (1) Microsoft Power BI: For interactive data visualization
- (2) Tableau: Known for robust, intuitive analytics and dashboards
- (3) Qlik: Focus on data integration and analytics
- (4) SAP Business Objects: A comprehensive suite for enterprise reporting.

36) Machine Learning:

37) Data and AI are transforming future careers by automating routine tasks, creating new roles, and enhancing human productivity.

Example

- (1) Customer Service: AI chatbots and voice assistants manage initial customer inquiries, providing 24/7 support and only escalating complex issues to human agents.
- (2) Manufacturing: Robotic systems perform inspection and assembly, decreasing human risk while increasing production speed.

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